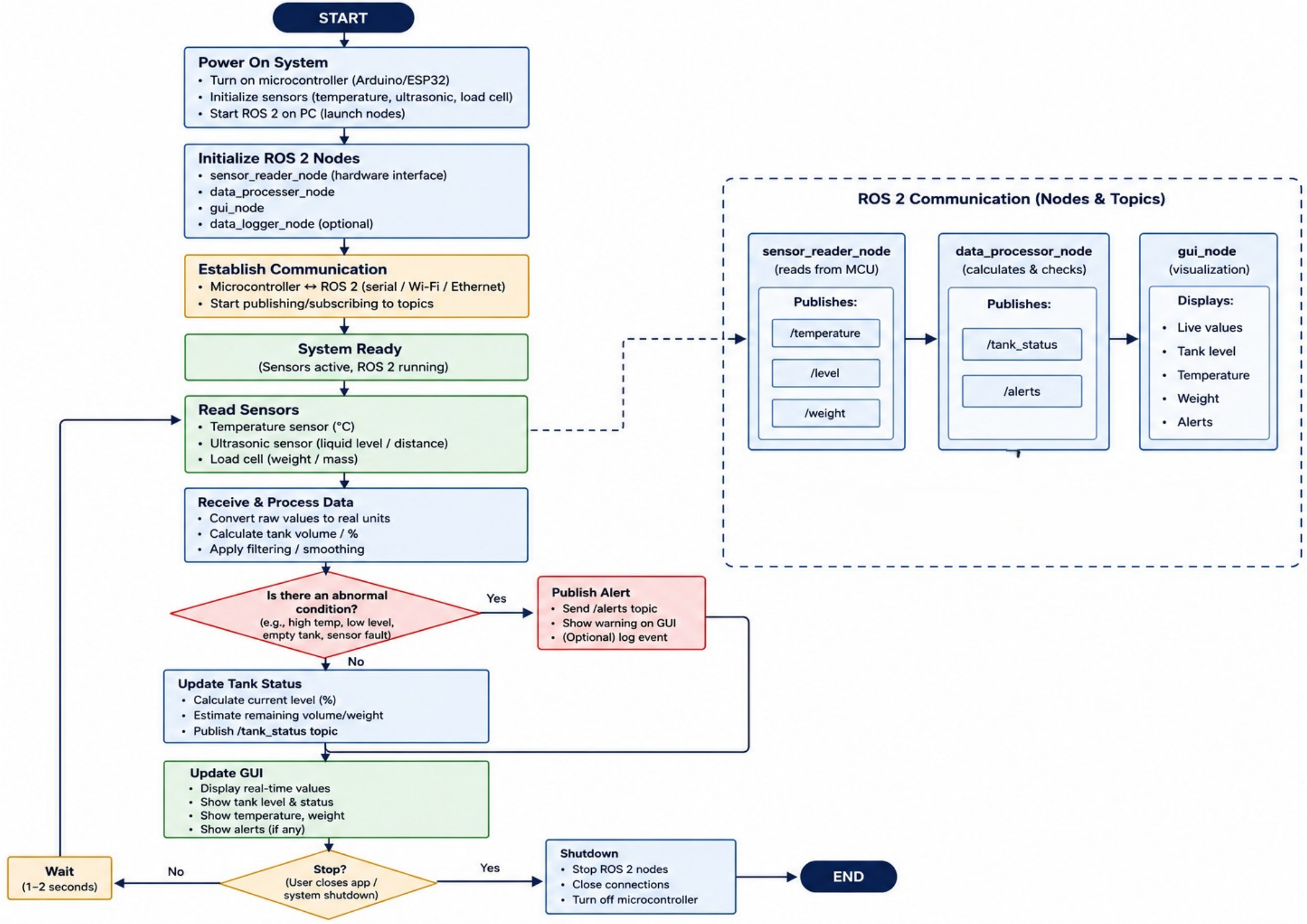

START
Power On System
• Turn on microcontroller (Arduino/ESP32)
• Initialize sensors (temperature, ultrasonic, load cell)
• Start ROS 2 on PC (launch nodes)
Initialize ROS 2 Nodes
• sensor_reader_node (hardware interface)
• data_processer_node
• gui_node
• data_logger_node (optional)
Establish Communication
• Microcontroller ↔ ROS 2 (serial / Wi-Fi / Ethernet)
• Start publishing/subscribing to topics
System Ready
(Sensors active, ROS 2 running)
Read Sensors
• Temperature sensor (°C)
• Ultrasonic sensor (liquid level / distance)
• Load cell (weight / mass)
Receive & Process Data
• Convert raw values to real units
• Calculate tank volume / %
• Apply filtering / smoothing
Is there an abnormal condition?
(e.g., high temp, low level, empty tank, sensor fault)
Yes
Publish Alert
• Send /alerts topic
• Show warning on GUI
• (Optional) log event
No
Update Tank Status
• Calculate current level (%)
• Estimate remaining volume/weight
• Publish /tank_status topic
Update GUI
• Display real-time values
• Show tank level & status
• Show temperature, weight
• Show alerts (if any)
Stop?
(User closes app / system shutdown)
No
Wait
(1–2 seconds)
Yes
Shutdown
• Stop ROS 2 nodes
• Close connections
• Turn off microcontroller
END
ROS 2 Communication (Nodes & Topics)
sensor_reader_node
(reads from MCU)
Publishes:
/temperature
/level
/weight
data_processor_node
(calculates & checks)
Publishes:
/tank_status
/alerts
gui_node
(visualization)
Displays:
• Live values
• Tank level
• Temperature
• Weight
• Alerts